Technical Note

PlatoSim Reference Frames

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Document History

Issue	Rev.	Date	Change description
1	0	2015-07-13	Initial version.
1	1	2015-08-27	Corrected gnomonic projection. Extended Section 3.
1	2	2015-09-01	Sign corrections, so that all rotations are now clockwise.
1	3	2016-02-11	Wrote the section about yaw, pitch, and roll.
2	1	2016-02-18	New relation between spacecraft and focal plane reference frames.
3	1	2016-04-17	Adapted spacecraft reference system to the one in [AD1].
4	1	2016-05-24	Added a small section the telescope drift rotation angles.
5	1	2016-06-22	Adapted the focal plane reference frame to the one of [AD2].
6	1	2017-01-17	Homogenised w.r.t. reference frames in updated [AD2].
6	2	2017-05-29	Changed CCD (A,B,C,D) to (1,2,3,4) nomenclature.
6	3	2017-09-18	Corrected values for the orientation angle for CCDs 2 and 4.
6	4	2020-02-16	Added the position of the platform pointing axis in the telescope reference frame
7	1	2020-04-29	Changed CCD reference frame definition. Extended Appendix B. Added figures.

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1 Abstract

2 Documents and Acronyms

2.1 Applicable Documents

AD1: PLATO-OHB-PL-LI-009, *PLATO Instrument Goordinate Systems*

AD2: PLATO-DLR-PL-TN-016, *PLATO Geometric Camera Model*

2.2 Reference Documents

RD1: P. Marcos-Arenal et al., 2014, A&A 566, A92, *The PLATO Simulator: Modelling of High-Cadence Space-Based Imaging*

2.3 Acronyms

CCD	Charge-Coupled Device
EQ	Equatorial reference frame
FP	Focal Plane
NP	Celestial Equatorial North Pole
PSF	Point Spread Function
SC	Spacecraft
SVM	Service Module
TL	Telescope (= Camera)

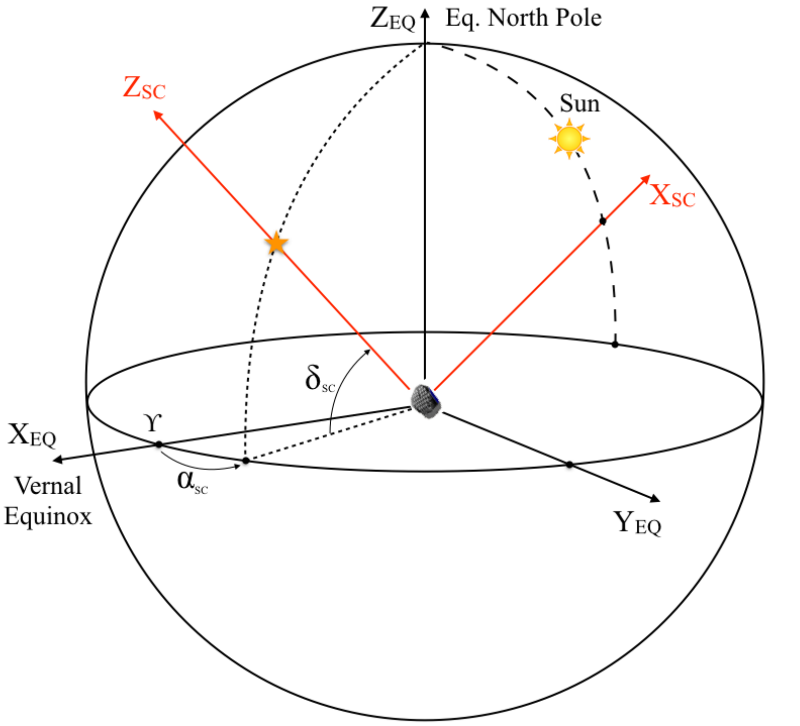

Figure 1: The base celestial equatorial coordinate system $(X_{\rm EQ}, Y_{\rm EQ}, Z_{\rm EQ})$. The positive $Z_{\rm SC}$ axis points towards the operator-given platform pointing coordinates $(\alpha_{\rm SC}, \delta_{\rm SC})$.

3 The Spacecraft reference frame

The base reference system of PlatoSim is the standard celestial equatorial coordinate system $(X_{\rm EQ}, Y_{\rm EQ}, Z_{\rm EQ})$ where $X_{\rm EQ}$ points towards the vernal equinox, and $Z_{\rm EQ}$ to the equatorial North Pole, as shown in Fig. 1.

The right-handed cartesian coordinate system $(X_{\rm SC}, Y_{\rm SC}, Z_{\rm SC})$, associated with the spacecraft, is the same as the one defined in [AD1]. Its origin is the geometric center of the interface between the bottom of the optical bench and the service module. The positive roll axis $Z_{\rm SC}$ points towards the operator-given mean payload line-of-sight, given by the equatorial sky coordinates $(\alpha_{\rm SC}, \delta_{\rm SC})$. The coordinates of the unit vector $\hat{\mathbf{z}}_{\rm SC}$ corresponding with the principal axis $Z_{\rm SC}$ can be computed in the equatorial reference frame with

$$\begin{aligned}\hat{\mathbf{z}}_{\rm SC} &= f(1, \pi/2 - \delta_{\rm SC}, \alpha_{\rm SC}) \\ &= (\cos\delta_{\rm SC}\cos\alpha_{\rm SC}, \cos\delta_{\rm SC}\sin\alpha_{\rm SC}, \sin\delta_{\rm SC})^t \end{aligned} \quad (1)$$

where f is the standard spherical coordinate transformation (22). The $X_{\rm SC}$ axis points in the direction of the highest point of the sunshield, as shown in Fig. 2. The spacecraft is rotated around its $Z_{\rm SC}$ axis, such that the $X_{\rm SC}$ axis intersects the great circle going through the celestial poles and the Sun. The coordinates of the corresponding unit vector $\hat{\mathbf{x}}_{\rm SC}$ in the equatorial reference frame are

$$\hat{\mathbf{x}}_{\rm SC} = (\cos\delta_x \cos\alpha_\odot, \cos\delta_x \sin\alpha_\odot, \sin\delta_x)^t \quad (2)$$

where $(\alpha_\odot, \delta_\odot)$ are the (mean) celestial coordinates of the Sun, and where the orthogonality requirement $\hat{\mathbf{x}}_{\rm SC} \cdot \hat{\mathbf{z}}_{\rm SC} = 0$ leads to the following implicit definition of δ_x:

$$\tan\delta_x = -\cot\delta_{\rm SC}\cos(\alpha_{\rm SC} - \alpha_\odot) \quad (3)$$

The unit vector $\hat{\mathbf{y}}_{\rm SC}$ completes the orthonormal reference frame:

$$\hat{\mathbf{y}}_{\rm SC} = \hat{\mathbf{z}}_{\rm SC} \times \hat{\mathbf{x}}_{\rm SC} \quad (4)$$

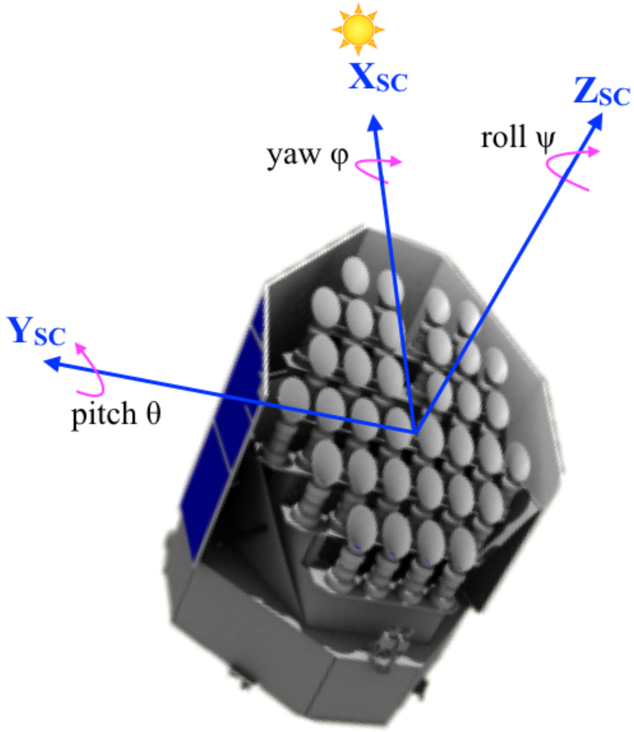

Figure 2: Definition of the right-handed $(X_{\rm SC}, Y_{\rm SC}, Z_{\rm SC})$ coordinate system. The positive $Z_{\rm SC}$ axis points towards the operator-given pointing coordinates $(\alpha_{\rm SC}, \delta_{\rm SC})$. The $X_{\rm SC}$ axis points in the direction of the highest point of the sunshield.

Given a target with celestial equatorial sky coordinates (α, δ), its cartesian coordinates $(v_x, v_y, v_z)_{\rm SC}$ in the spacecraft reference frame are computed using the following transformation:

$$\begin{pmatrix} v_x \\ v_y \\ v_z \end{pmatrix}_{\rm SC} = \mathbf{R}^{\rm SC}_{\rm EQ}(\alpha_{\rm SC}, \delta_{\rm SC}, \alpha_\odot, \delta_\odot) \begin{pmatrix} \cos\delta\cos\alpha \\ \cos\delta\sin\alpha \\ \sin\delta \end{pmatrix}_{\rm EQ} \tag{5}$$

where the rotation matrix $\mathbf{R}^{\rm SC}_{\rm EQ}sc(\alpha_{\rm SC}, \delta_{\rm SC}, \alpha_\odot, \delta_\odot)$ is given by

$$\mathbf{R}^{\rm SC}_{\rm EQ}(\alpha_{\rm SC}, \delta_{\rm SC}, \alpha_\odot, \delta_\odot) = \begin{pmatrix} \hat{\mathbf{x}}_{{\rm SC},x} & \hat{\mathbf{x}}_{{\rm SC},y} & \hat{\mathbf{x}}_{{\rm SC},z} \\ \hat{\mathbf{y}}_{{\rm SC},x} & \hat{\mathbf{y}}_{{\rm SC},y} & \hat{\mathbf{y}}_{{\rm SC},z} \\ \hat{\mathbf{z}}_{{\rm SC},x} & \hat{\mathbf{z}}_{{\rm SC},y} & \hat{\mathbf{z}}_{{\rm SC},z} \end{pmatrix} \tag{6}$$

where the equatorial cartesian coordinates of the unit vectors $\hat{\mathbf{x}}_{\rm SC}$, $\hat{\mathbf{y}}_{\rm SC}$, and $\hat{\mathbf{z}}_{\rm SC}$ are given by Eqs. (1), (2), and (4).

4 The telescope reference frame

For historical reasons, throughout its code PlatoSim uses a somewhat different definition of "telescope" than used elsewhere in the Plato consortium. Telescope here is definined as the combined unit of baffle, optical elements, and detector. Elsewhere this would be called the "camera".

The right-handed cartesian coordinate system $(X_{\rm TL}, Y_{\rm TL}, Z_{\rm TL})$ is associated with one of the telescopes and has its origin exactly in the middle of the 4 CCDs in the associated focal plane. The $Z_{\rm TL}$ axis is pointing towards the field-of-view and is called the optical axis, which is usually not parallell with the spacecraft pointing axis

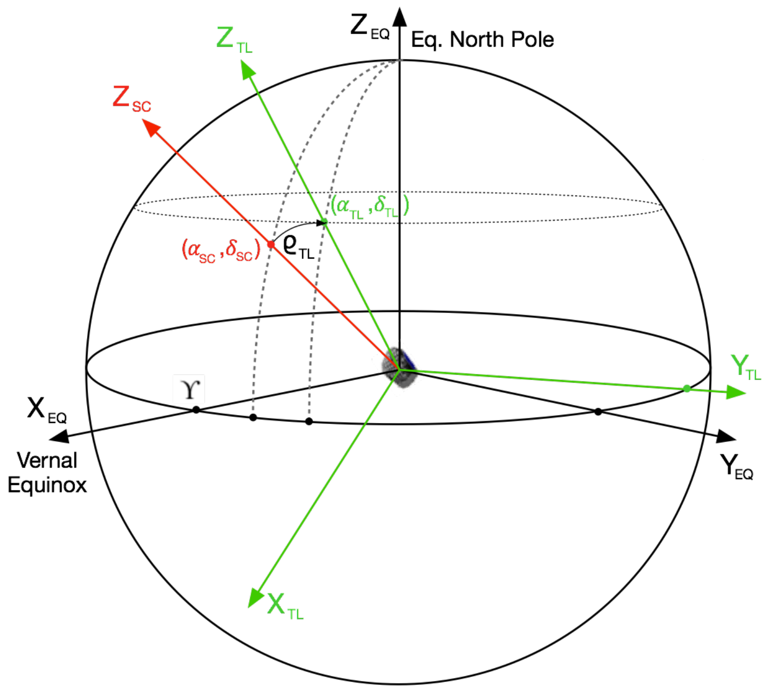

Figure 3: Definition of the Telescope reference frame $(X_{\rm TL}, Y_{\rm TL}, Z_{\rm TL})$. The focal plane is defined by the plane $(X_{\rm TL}, Y_{\rm TL})$, where the axis $Y_{\rm TL}$ is in the equatorial plane. $Z_{\rm TL}$ is the optical axis of the telescope, pointing towards the field-of-view.

$Z_{\rm SC}$ as shown in Fig. 3. To specify how the telescope optical axis is oriented with respect to the platform, two (user-given) angles are used: the tilt angle $\varrho_{\rm TL}$ between $Z_{\rm TL}$ and $Z_{\rm SC}$, and the position angle $\eta_{\rm TL}$ of a rotation around the positive $Z_{\rm SC}$ axis. The orientation of the unit vector $\hat{\mathbf{z}}_{\rm TL}$ corresponding to the optical axis $Z_{\rm TL}$, can be computed by first rotating the $(X_{\rm SC}, Y_{\rm SC})$ plane around the pointing axis $Z_{\rm SC}$ over an angle $\eta_{\rm TL}$ to get the reference frame $(X'_{\rm SC}, Y'_{\rm SC}, Z'_{\rm SC})$, then rotating the $Z'_{\rm SC} = Z_{\rm SC}$ axis over an angle $\rho_{\rm TL}$ around the $Y'_{\rm SC}$ axis to get the $(X''_{\rm SC}, Y''_{\rm SC}, Z''_{\rm SC})$ reference frame, and finally rotate the $(X''_{\rm SC}, Y''_{\rm SC})$ plane around the $Z''_{\rm SC}$ axis over an angle $2\pi - \eta_{\rm TL} = -\eta_{\rm TL}$ to obtain the telescope reference frame. These rotations are illustrated in Fig. 4.

A vector with coordinates $(x_{\rm SC}, y_{\rm SC}, z_{\rm SC})$ in the spacecraft reference frame, has coordinates $(x_{\rm TL}, y_{\rm TL}, z_{\rm TL})$ in the telescope reference frame that can be computed using the following transformation:

$$\begin{pmatrix} x_{\rm TL} \\ y_{\rm TL} \\ z_{\rm TL} \end{pmatrix} = \mathbf{R}_{\rm SC}^{\rm TL}(\eta_{\rm TL}, \rho_{\rm TL}) \begin{pmatrix} x_{\rm SC} \\ y_{\rm SC} \\ z_{\rm SC} \end{pmatrix} \tag{7}$$

where

$$\mathbf{R}_{\rm SC}^{\rm TL}(\eta_{\rm TL}, \rho_{\rm TL}) = \begin{pmatrix} \cos\eta_{\rm TL} & -\sin\eta_{\rm TL} & 0 \\ \sin\eta_{\rm TL} & \cos\eta_{\rm TL} & 0 \\ 0 & 0 & 1 \end{pmatrix} \begin{pmatrix} \cos\varrho_{\rm TL} & 0 & -\sin\varrho_{\rm TL} \\ 0 & 1 & 0 \\ \sin\varrho_{\rm TL} & 0 & \cos\varrho_{\rm TL} \end{pmatrix} \begin{pmatrix} \cos\eta_{\rm TL} & \sin\eta_{\rm TL} & 0 \\ -\sin\eta_{\rm TL} & \cos\eta_{\rm TL} & 0 \\ 0 & 0 & 1 \end{pmatrix} \tag{8}$$

where we used the rotation matrices defined in Appendix B.2. The inverse transformation is:

$$\begin{pmatrix} x_{\rm SC} \\ y_{\rm SC} \\ z_{\rm SC} \end{pmatrix} = \mathbf{R}_{\rm TL}^{\rm SC}(\eta_{\rm TL}, \rho_{\rm TL}) \begin{pmatrix} x_{\rm TL} \\ y_{\rm TL} \\ z_{\rm TL} \end{pmatrix} \tag{9}$$

where

$$\mathbf{R}_{\rm TL}^{\rm SC}(\eta_{\rm TL}, \rho_{\rm TL}) = \begin{pmatrix} \cos\eta_{\rm TL} & -\sin\eta_{\rm TL} & 0 \\ \sin\eta_{\rm TL} & \cos\eta_{\rm TL} & 0 \\ 0 & 0 & 1 \end{pmatrix} \begin{pmatrix} \cos\varrho_{\rm TL} & 0 & \sin\varrho_{\rm TL} \\ 0 & 1 & 0 \\ -\sin\varrho_{\rm TL} & 0 & \cos\varrho_{\rm TL} \end{pmatrix} \begin{pmatrix} \cos\eta_{\rm TL} & \sin\eta_{\rm TL} & 0 \\ -\sin\eta_{\rm TL} & \cos\eta_{\rm TL} & 0 \\ 0 & 0 & 1 \end{pmatrix} \tag{10}$$

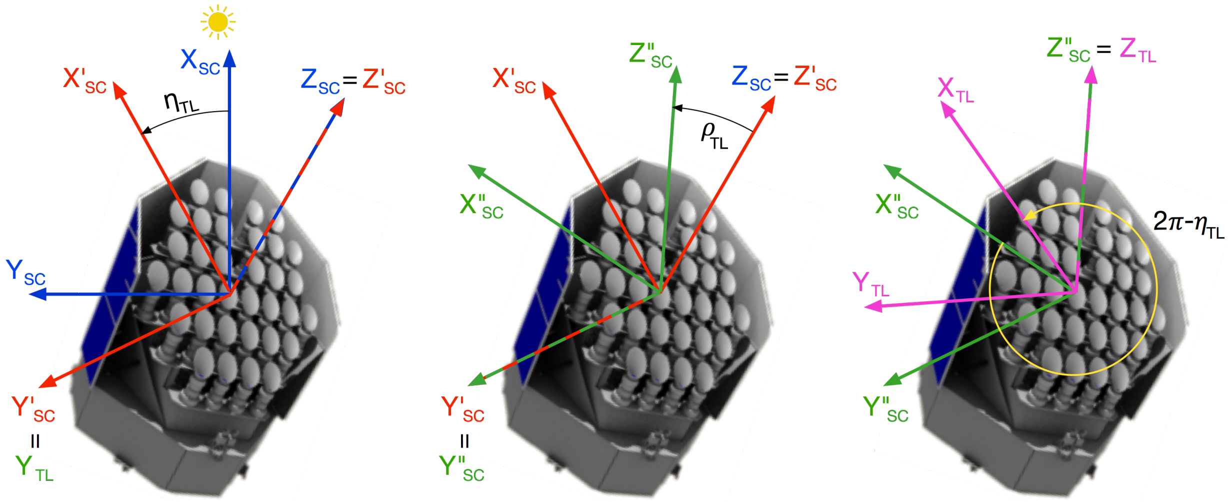

Figure 4: The telescope reference frame can be obtained from the spacecraft reference frame by first rotating the $(X_{\mathrm{SC}}, Y_{\mathrm{SC}})$ plane around the pointing axis Z_{SC} over an angle η_{TL}, then rotating the resulting Z'_{SC} axis over a tilt angle ρ_{TL} around the Y'_{SC} axis, and finally rotating over an angle $2\pi - \eta_{\mathrm{TL}} = -\eta_{\mathrm{TL}}$ around the Z''_{SC} axis.

is the transpose of the rotation matrix $\mathbf{R}_{\mathrm{SC}}^{\mathrm{TL}}(\eta_{\mathrm{TL}}, \rho_{\mathrm{TL}})$. The angles η_{TL} and ρ_{TL} for the different camera groups of PlatoSim are listed in Table 1. The tilt angle ρ_{TL} is fixed at 9.2 degrees, and the azimuth angle is $\eta_{\mathrm{TL}} = 45 + n \cdot 90$ degrees where n is the camera group ID number for the normal cameras.

Camera Group	η_{TL} (°)	ρ_{TL} (°)
1	45	9.2
2	135	9.2
3	225	9.2
4	315	9.2
Fast	0	0.0

Table 1: The first column lists the camera group ID number, the second column the azimuth rotation anlge η_{TL} in degrees, and the third column the tilt rotation angle ρ_{TL} in degrees.

To understand why it is useful to make an extra rotation over the angle $-\eta_{\mathrm{TL}}$ after rotating first over the angles η_{TL} and ρ_{TL}, we first observe that *without* the extra rotation, the location of the spacecraft pointing axis $(x_{\mathrm{SC}}, y_{\mathrm{SC}}, z_{\mathrm{SC}}) = (0, 0, 1)$ in the telescope reference frame would be $(x_{\mathrm{TL}}, y_{\mathrm{TL}}, z_{\mathrm{TL}}) = (-\sin\varrho, 0, \cos\varrho)$ (cf. Eq. (7)). That is, in the telescope xy-plane the spacecraft pointing axis would cross the negative x_{TL}-axis, independent of the camera group. This is illustrated in Fig. 5. The extra rotation allows to align the CCDs of the different camera groups in the sky, and has the additional benefit that the platform pointing coordinates always fall on (rather than in between) a CCD for each of the 4 camera groups.

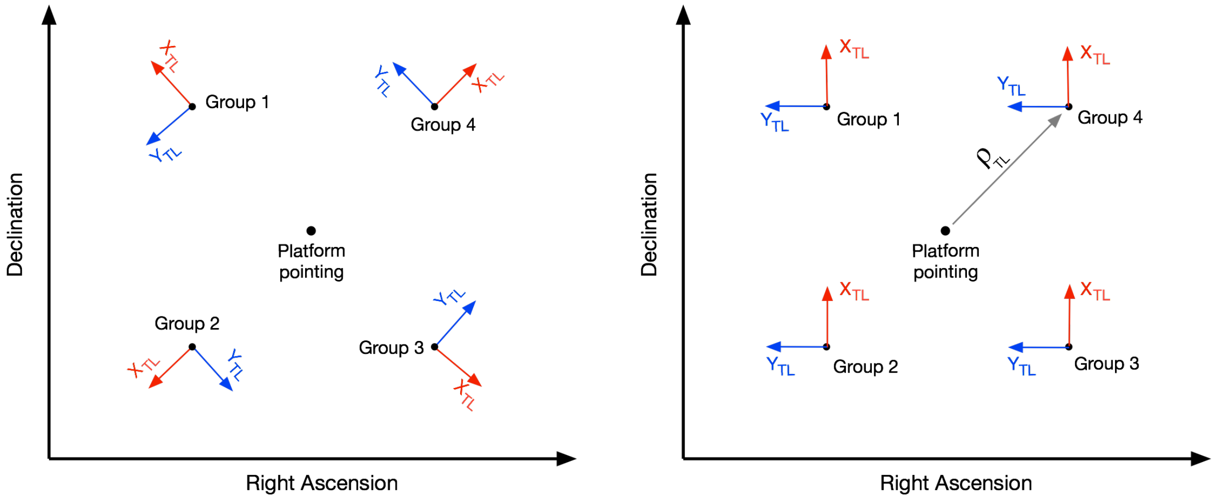

Figure 5: Left: the orientation of the telescope reference frame for each of the camera groups when there would be not extra rotation over an angle $-\eta_{\rm TL}$. Right: the orientation of the same reference frames with the extra rotation as outlined in Eq. (8). $\rho_{\rm TL}$ is the camera tilt angle. The righthand side orientation allows for nearly aligned CCDs of the different camera groups on the sky.

5 The Focal Plane reference frame

The right-handed cartesian coordinate system $(X_{\rm FP}, Y_{\rm FP}, Z_{\rm FP})$ is associated with the Focal Plane of one camera, and like the telescope reference frame it has its origin exactly in the middle of the 4 CCDs. The $Z_{\rm FP}$ axis coincides with the $Z_{\rm TL}$ optical axis of the telescope which points towards the field-of-view. The focal plane $(X_{\rm FP}, Y_{\rm FP})$ is obtained by rotating the $(X_{\rm TL}, Y_{\rm TL})$ plane over an angle $\gamma_{\rm FP}$ called the focal plane orientation around the $Z_{\rm FP} = Z_{\rm TL}$ axis. This is illustrated in Fig. 6. The corresponding tranformation (using homogeneous coordinates, see Appendix B.3) is:

$$\begin{pmatrix} x_{\rm FP} \\ y_{\rm FP} \\ f \\ 1 \end{pmatrix} = \mathbf{R}_{\rm TL}^{\rm FP}(\gamma_{\rm FP}) \begin{pmatrix} x_{\rm TL} \\ y_{\rm TL} \\ z_{\rm TL} \\ 1 \end{pmatrix} \tag{11}$$

where the rotation matrix $\mathbf{R}_{\rm TL}^{\rm FP}(\gamma_{\rm FP})$ is given by:

$$\mathbf{R}_{\rm TL}^{\rm FP}(\gamma_{\rm FP}) = \begin{pmatrix} f & 0 & 0 & 0 \\ 0 & f & 0 & 0 \\ 0 & 0 & f & 0 \\ 0 & 0 & 1 & 0 \end{pmatrix} \begin{pmatrix} \cos\gamma_{\rm FP} & \sin\gamma_{\rm FP} & 0 & 0 \\ -\sin\gamma_{\rm FP} & \cos\gamma_{\rm FP} & 0 & 0 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 \end{pmatrix} \tag{12}$$

Here $(x_{\rm TL}, y_{\rm TL}, z_{\rm TL}, 1)$ are the homogenized coordinates of a vector in the telescope reference frame, and $(x_{\rm FP}, y_{\rm FP}, f, 1)$ are the coordinates of the homogeneous projected vector in the focal plane reference frame. Note that $\mathbf{R}_{\rm TL}^{\rm FP}$ is not invertible: projecting loses the $z_{\rm TL}$ information. However, since PlatoSim only uses (sky) directions, the reverse transformation can be obtained by enforcing the length of the vector in the telescope reference frame to be unity.

6 The CCD reference frame

The CCD coordinate system $(X_{\rm CCD}, Y_{\rm CCD})$ is associated with 1 CCD, and has its origin in one of the corners of the CCD. The charge transfer direction during readout of the CCD happens in the negative $Y_{\rm CCD}$ direction towards $y_{\rm CCD} = 0$, and the readout of the serial readout register happens in the negative $X_{\rm CCD}$ direction towards the origin. The orientation of the CCD with respect to the $(X_{\rm FP}, Y_{\rm FP})$ coordinate system is defined by the angle $\gamma_{\rm CCD}$, as shown in Fig. 6.

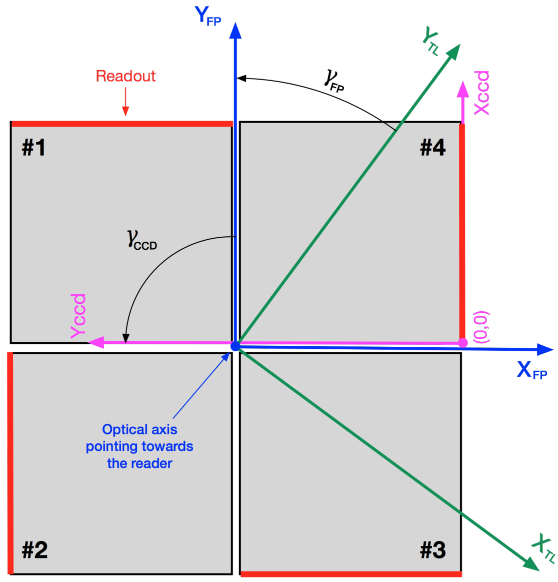

Figure 6: A schematic overview of the focal plane with 4 CCDs. The optical axis $Z_{\rm FP}$ is the blue dot in the middle of the 4 CCDs and points in the positive direction towards the reader. The green axes represent the $(x_{\rm TL}, y_{\rm TL})$ plane in the telescope reference frame. $\gamma_{\rm CCD}$ is the CCD rotation angle.

The transformation (in homogeneous coordinates) between the focal plane and the CCD reference frames is implemented with

$$\begin{pmatrix} x_{\rm CCD} \\ y_{\rm CCD} \\ 1 \end{pmatrix} = \mathbf{R}_{\rm FP}^{\rm CCD}(\gamma_{\rm CCD}) \begin{pmatrix} x_{\rm FP} \\ y_{\rm FP} \\ 1 \end{pmatrix} \tag{13}$$

where

$$\mathbf{R}_{\rm FP}^{\rm CCD}(\gamma_{\rm CCD}) = \begin{pmatrix} \cos\gamma_{\rm CCD} & \sin\gamma_{\rm CCD} & \Delta x_{\rm CCD} \\ -\sin\gamma_{\rm CCD} & \cos\gamma_{\rm CCD} & \Delta y_{\rm CCD} \\ 0 & 0 & 1 \end{pmatrix} \tag{14}$$

where $(\Delta x_{\rm CCD}, \Delta y_{\rm CCD})$ are the coordinates of the CCD zeropoint in the $(X_{\rm FP}, Y_{\rm FP})$ frame. Note that although we use the same rotation matrix as in Eq. (39), we deviated with the placement of the translation vector for practical reasons. Eq. (13) allows to use the same $\Delta x_{\rm CCD}$ and $\Delta y_{\rm CCD}$ for each CCD, independent of the CCD rotation angle $\gamma_{\rm CCD}$. An overview of the zero point coordinates and the orientation angle $\gamma_{\rm CCD}$ is given in Table 2. PlatoSim chooses to define the Focal Plane reference frame such that its axes run parallel with the CCD axes of CCD #3.

CCD	$N_{\rm rows}$	$N_{\rm cols}$	$r_{\rm exp}$	$\Delta x_{\rm CCD}$ (mm)	$\Delta y_{\rm CCD}$ (mm)	$\gamma_{\rm CCD}$ (deg)
1	4510	4510	0	-1.3	82.48	180
2	4510	4510	0	-1.3	82.48	270
3	4510	4510	0	-1.3	82.48	0
4	4510	4510	0	-1.3	82.48	90
1 (F)	4510	4510	2255	-1.3	82.48	180
2 (F)	4510	4510	2255	-1.3	82.48	270
3 (F)	4510	4510	2255	-1.3	82.48	0
4 (F)	4510	4510	2255	-1.3	82.48	90

Table 2: The first column lists the CCD codes: 1,2,3,4 for a normal camera, 1 (F), 2 (F), 3 (F), 4 (F) for a fast camera. $r_{\rm exp}$ is the first row that is exposed; for the fast camera only rows 2255 until 4510 are exposed. $\Delta x_{\rm CCD}$ and $\Delta y_{\rm CCD}$ are the CCD origin coordinates in the $(X_{\rm FP}, Y_{\rm FP})$ frame. $\gamma_{\rm CCD}$ is the CCD rotation angle.

7 Radial dependency of the PSF

In realistic cases, the PSF is dependent on the angular radial distance θ between the projected center $(x_{\rm FP}, y_{\rm FP})$ of the PSF in the focal plane to the optical axis which is assumed to go through the origin $(0,0)$ of the focal plane. The cosine of this angular distance can be computed using

$$\cos\theta = \frac{\mathbf{v}\cdot\mathbf{z}}{\|\mathbf{v}\|\ \|\mathbf{z}\|} \tag{15}$$

where $\mathbf{v} \equiv (x_{\rm FP}, y_{\rm FP}, -f)$ is the coordinate vector of the PSF in the pinhole reference frame (see Fig. 9), and $\mathbf{z} = (0,0,-1)$ is the unit vector along the optical axis but pointing towards the focal plane, also in the pinhole reference frame. This leads to

$$\cos\theta = \left[1 + \left(\frac{x_{\rm FP}}{f}\right)^2 + \left(\frac{y_{\rm FP}}{f}\right)^2\right]^{-1/2} \tag{16}$$

where $(x_{\rm FP}, y_{\rm FP})$ are the pinhole-projected coordinates of the center of the PSF, and where f is the focal length of the camera.

8 Platform jitter rotation angles

We define the platform yaw φ, the pitch θ, and the roll ψ as rotation angles around respectively the $X_{\rm SC}$, $Y_{\rm SC}$, and $Z_{\rm SC}$ axes, such that the angles increase with a clockwise rotation, when looking along the positive axes. This is illustrated in Fig. 2.

First a roll rotation is done around the $Z_{\rm SC}$ axis, then a pitch rotation is done around the rotated $Y_{\rm SC}$ axis, and finally a yaw rotation is done around the twice-rotated $X_{\rm SC}$ axis. The combined rotation matrix (in the Spacecraft SC reference frame) is given by (cf. Eq. (27)):

$$R(\theta,\varphi,\psi) = R(\theta)R(\phi)R(\psi) \tag{17}$$

$$= \begin{pmatrix} 1 & 0 & 0 \\ 0 & \cos\varphi & -\sin\varphi \\ 0 & \sin\varphi & \cos\varphi \end{pmatrix} \begin{pmatrix} \cos\theta & 0 & \sin\theta \\ 0 & 1 & 0 \\ -\sin\theta & 0 & \cos\theta \end{pmatrix} \begin{pmatrix} \cos\psi & -\sin\psi & 0 \\ \sin\psi & \cos\psi & 0 \\ 0 & 0 & 1 \end{pmatrix} \tag{18}$$

with the inverse transformation

$$R^{-1}(\theta,\varphi,\psi) = R^t(\psi)R^t(\phi)R^t(\theta) \tag{19}$$

$$= \begin{pmatrix} \cos\psi & \sin\psi & 0 \\ -\sin\psi & \cos\psi & 0 \\ 0 & 0 & 1 \end{pmatrix} \begin{pmatrix} \cos\theta & 0 & -\sin\theta \\ 0 & 1 & 0 \\ \sin\theta & 0 & \cos\theta \end{pmatrix} \begin{pmatrix} 1 & 0 & 0 \\ 0 & \cos\varphi & \sin\varphi \\ 0 & -\sin\varphi & \cos\varphi \end{pmatrix} \tag{20}$$

To compute the updated equatorial sky coordinates $(\tilde{\alpha}_{\rm SC}, \tilde{\delta}_{\rm SC})$ of the platform Z axis after jitter, we first apply the yaw, jitter, and roll rotations to the current roll axis $\hat{\mathbf{z}}_{\rm SC} = (0,0,1)$ in the spacecraft reference frame:

$$\tilde{\mathbf{z}}_{\rm SC} = R(\theta,\varphi,\psi)\cdot\hat{\mathbf{z}}_{\rm SC} \tag{21}$$

and then use the inverse transformation of Eq. (5) and Eq. 23 to compute the updated $(\tilde{\alpha}_{\rm SC}, \tilde{\delta}_{\rm SC})$.

Acknowledgements

We thank Denis Grießbach of DLR for helpful comments.

A Spherical Coordinate Transformation

We use the following notation for the standard 3D spherical coordinate transformation:

$$f : \mathbb{R}^+ \times [0,\pi] \times [0,2\pi] \to \mathbb{R}^3 : \begin{pmatrix} r \\ \theta \\ \phi \end{pmatrix} \mapsto \begin{pmatrix} x \\ y \\ z \end{pmatrix} \equiv \begin{pmatrix} r\sin\theta\cos\phi \\ r\sin\theta\sin\phi \\ r\cos\theta \end{pmatrix} \tag{22}$$

and its inverse:

$$f^{-1} : \mathbb{R}^3 \to \mathbb{R}^+ \times [0,\pi] \times [0,2\pi] : \begin{pmatrix} x \\ y \\ z \end{pmatrix} \mapsto \begin{pmatrix} r \\ \theta \\ \phi \end{pmatrix} \equiv \begin{pmatrix} \sqrt{x^2+y^2+z^2} \\ \arccos\left(z/\sqrt{x^2+y^2+z^2}\right) \\ \arctan(y/x) \end{pmatrix} \tag{23}$$

B Coordinate transformations

B.1 The components of a rotated vector in the same reference frame

The coordinate transformation in this Section is sometimes called in the literature an *active transformation* or an *alibi transformation*. We consider a vector $\mathbf{v} = v_x\mathbf{e}_x + v_y\mathbf{e}_y + v_z\mathbf{e}_z$ that is first rotated by an angle θ around an axis determined by the unit vector $\hat{\mathbf{u}} = u_x\mathbf{e}_x + u_y\mathbf{e}_y + u_z\mathbf{e}_z$, and then translated by a vector $\mathbf{t} = t_x\mathbf{e}_x + t_y\mathbf{e}_y + t_z\mathbf{e}_z$ to give a vector $\mathbf{v}' = v'_x\mathbf{e}_x + v'_y\mathbf{e}_y + v'_z\mathbf{e}_z$. An example is given in Fig. 7 where the rotation axis $\mathbf{u} = \mathbf{e}_z$ is pointing towards the reader. The components of the two vectors are related by:

$$\begin{pmatrix} v'_x \\ v'_y \\ v'_z \end{pmatrix} = R_{\mathbf{u}}(\theta) \begin{pmatrix} v_x \\ v_y \\ v_z \end{pmatrix} + \begin{pmatrix} t_x \\ t_y \\ t_z \end{pmatrix} \tag{24}$$

and the inverse transformation

$$\begin{pmatrix} v_x \\ v_y \\ v_z \end{pmatrix} = R^t_{\mathbf{u}}(\theta) \begin{pmatrix} v'_x \\ v'_y \\ v'_z \end{pmatrix} - R^t_{\mathbf{u}}(\theta) \begin{pmatrix} t_x \\ t_y \\ t_z \end{pmatrix} = R^t_{\mathbf{u}}(\theta) \begin{pmatrix} v'_x - t_x \\ v'_y - t_y \\ v'_z - t_z \end{pmatrix} \tag{25}$$

where the components of the rotation matrix $R_{\mathbf{u}}(\theta)$ are given by the *Rodrigues formula*:

$$R_{ij}(\mathbf{u},\theta) = \cos\theta\ \delta_{ij} + (1-\cos\theta)u_iu_j - \sin\theta\epsilon_{ijk}u_k. \tag{26}$$

Or, written in full:

$$R_{\mathbf{u}}(\theta) = \begin{pmatrix} \cos\theta + u_x^2(1-\cos\theta) & u_xu_y(1-\cos\theta) - u_z\sin\theta & u_xu_z(1-\cos\theta) + u_y\sin\theta \\ u_yu_x(1-\cos\theta) + u_z\sin\theta & \cos\theta + u_y^2(1-\cos\theta) & u_yu_z(1-\cos\theta) - u_x\sin\theta \\ u_zu_x(1-\cos\theta) - u_y\sin\theta & u_zu_y(1-\cos\theta) + u_x\sin\theta & \cos\theta + u_z^2(1-\cos\theta) \end{pmatrix}. \tag{27}$$

$R^t_{\mathbf{u}}(\theta) = R^{-1}_{\mathbf{u}}(\theta)$ denotes the transposed rotation matrix. Looking into the positive direction of $\mathbf{u}$ the rotation angle θ increases clockwise (right-hand rule). It is readily seen that $R_{-\mathbf{u}}(-\theta) = R_{\mathbf{u}}(\theta)$. For the rotation matrices around the principle axes $\mathbf{e}_x$, $\mathbf{e}_y$, and $\mathbf{e}_z$, we obtain

$$R_{\mathbf{e}_x}(\theta) = \begin{pmatrix} 1 & 0 & 0 \\ 0 & \cos\theta & -\sin\theta \\ 0 & \sin\theta & \cos\theta \end{pmatrix}, \tag{28}$$

$$R_{\mathbf{e}_y}(\theta) = \begin{pmatrix} \cos\theta & 0 & \sin\theta \\ 0 & 1 & 0 \\ -\sin\theta & 0 & \cos\theta \end{pmatrix}, \tag{29}$$

and

$$R_{\mathbf{e}_z}(\theta) = \begin{pmatrix} \cos\theta & -\sin\theta & 0 \\ \sin\theta & \cos\theta & 0 \\ 0 & 0 & 1 \end{pmatrix}. \tag{30}$$

Note that the pattern of the minus signs for the case of $R_{\mathbf{e}_y}(\theta)$ differs from that of $R_{\mathbf{e}_z}(\theta)$ and $R_{\mathbf{e}_z}(\theta)$.

Using so-called *homogeneous* coordinates, we can combine the rotation and the translation in a single matrix multiplication:

$$\begin{pmatrix} v'_x \\ v'_y \\ v'_z \\ 1 \end{pmatrix} = \begin{pmatrix} R_{11} & R_{12} & R_{13} & t_x \\ R_{21} & R_{22} & R_{23} & t_y \\ R_{31} & R_{32} & R_{33} & t_z \\ 0 & 0 & 0 & 1 \end{pmatrix} \begin{pmatrix} v_x \\ v_y \\ v_z \\ 1 \end{pmatrix}, \tag{31}$$

and the same for the inverse transformation.

$$\begin{pmatrix} v_x \\ v_y \\ v_z \\ 1 \end{pmatrix} = \begin{pmatrix} R_{11} & R_{12} & R_{13} & t_x \\ R_{21} & R_{22} & R_{23} & t_y \\ R_{31} & R_{32} & R_{33} & t_z \\ 0 & 0 & 0 & 1 \end{pmatrix}^{-1} \begin{pmatrix} v'_x \\ v'_y \\ v'_z \\ 1 \end{pmatrix} = \left(\begin{array}{c|c} R^t_{\mathbf{u}}(\theta) & -R^t_{\mathbf{u}}(\theta)\,\mathbf{t} \\ \hline 0 & 1 \end{array}\right) \begin{pmatrix} v'_x \\ v'_y \\ v'_z \\ 1 \end{pmatrix}. \tag{32}$$

B.2 The components of a vector in a rotated reference frame

The coordinate transformation in this Section is sometimes called in the literature a *passive transformation* or an *alias transformation*. Given a first reference frame $(\mathbf{e}_x, \mathbf{e}_y, \mathbf{e}_z)$, we consider a rotated reference frame of which the principle unit vectors $(\mathbf{e}_{x'}, \mathbf{e}_{y'}, \mathbf{e}_{z'})$ are defined by rotating the principle unit vectors of the first frame, as described in the previous section:

$$(\mathbf{e}_{x'}, \mathbf{e}_{y'}, \mathbf{e}_{z'}) \equiv (R_{\mathbf{u}(\theta)}\mathbf{e}_x, R_{\mathbf{u}(\theta)}\mathbf{e}_y, R_{\mathbf{u}(\theta)}\mathbf{e}_z) \tag{33}$$

Here $R_{\mathbf{u}}(\theta)$ is a rotation matrix given by Exp. (27). The rotation angle θ increases clockwise when looking in the positive direction of the rotation vector u. An example is given in Fig. 8 where the rotation axis $\mathbf{u} = \mathbf{e}_z$ is pointing towards the reader. The components of a vector $\mathbf{v} = v_x\mathbf{e}_x + v_y\mathbf{e}_y + v_z\mathbf{e}_z = v_{x'}\mathbf{e}_{x'} + v_{y'}\mathbf{e}_{y'} + v_{z'}\mathbf{e}_{z'}$ in the two reference frames are related by

$$\begin{pmatrix} v_{x'} \\ v_{y'} \\ v_{z'} \end{pmatrix} = R^t_{\mathbf{u}}(\theta) \begin{pmatrix} v_x \\ v_y \\ v_z \end{pmatrix} \tag{34}$$

and

$$\begin{pmatrix} v_x \\ v_y \\ v_z \end{pmatrix} = R_{\mathbf{u}}(\theta) \begin{pmatrix} v_{x'} \\ v_{y'} \\ v_{z'} \end{pmatrix} \tag{35}$$

In case there is also a translation vector $\mathbf{t} = t_x\mathbf{e}_x + t_y\mathbf{e}_y + t_z\mathbf{e}_z$ involved, we opt to place this vector such that there is consistency with Eqs. (24)-(25).

$$\begin{pmatrix} v_{x'} \\ v_{y'} \\ v_{z'} \end{pmatrix} = R^t_{\mathbf{u}}(\theta) \begin{pmatrix} v_x \\ v_y \\ v_z \end{pmatrix} - R^t_{\mathbf{u}}(\theta) \begin{pmatrix} t_x \\ t_y \\ t_z \end{pmatrix} = R^t_{\mathbf{u}}(\theta) \begin{pmatrix} v_x - t_x \\ v_y - t_y \\ v_z - t_z \end{pmatrix} \tag{36}$$

and

$$\begin{pmatrix} v_x \\ v_y \\ v_z \end{pmatrix} = R_{\mathbf{u}}(\theta) \begin{pmatrix} v_{x'} \\ v_{y'} \\ v_{z'} \end{pmatrix} + \begin{pmatrix} t_x \\ t_y \\ t_z \end{pmatrix} \tag{37}$$

Using homogeneous coordinates we can write

$$\begin{pmatrix} v_x \\ v_y \\ v_z \\ 1 \end{pmatrix} = \begin{pmatrix} R_{11} & R_{12} & R_{13} & t_x \\ R_{21} & R_{22} & R_{23} & t_y \\ R_{31} & R_{32} & R_{33} & t_z \\ 0 & 0 & 0 & 1 \end{pmatrix} \begin{pmatrix} v_{x'} \\ v_{y'} \\ v_{z'} \\ 1 \end{pmatrix}, \tag{38}$$

and the inverse transformation.

$$\begin{pmatrix} v_{x'} \\ v_{y'} \\ v_{z'} \\ 1 \end{pmatrix} = \begin{pmatrix} R_{11} & R_{12} & R_{13} & t_x \\ R_{21} & R_{22} & R_{23} & t_y \\ R_{31} & R_{32} & R_{33} & t_z \\ 0 & 0 & 0 & 1 \end{pmatrix}^{-1} \begin{pmatrix} v_x \\ v_y \\ v_z \\ 1 \end{pmatrix} = \left(\begin{array}{c|c} R^t_{\mathbf{u}}(\theta) & -R^t_{\mathbf{u}}(\theta)\,\mathbf{t} \\ \hline 0 & 1 \end{array}\right) \begin{pmatrix} v_x \\ v_y \\ v_z \\ 1 \end{pmatrix}. \tag{39}$$

B.3 Pinhole projection

The standard pinhole camera model is shown in Fig. 9. The Z axis is the optical axis of the telescope, the (X, Y) plane is parallel to the focal plane, but has its origin in the pinhole. The focal plane (with the CCDs) is at a distance f behind the pinhole, where f is the focal length of the camera. Light rays of a star at point P go through the pin hole O and are projected to a point Q on the focal plane. In the (Y, Z) plane, the triangles $\widehat{OP'P''}$ and $\widehat{OQ'Q''}$ are similar triangles, so that we have $|P'P''|/|OP'| = |Q'Q''|/|OQ'|$ or

$$\frac{-y_q}{f} = \frac{y_p}{z_p} \tag{40}$$

where the minus sign indicates that the pinhole reverses the image on the focal plane. A similar argument holds in the (X, Z) plane, so that we can write

$$\begin{pmatrix} x_q \\ y_q \\ -f \end{pmatrix} = -\frac{f}{z_p} \begin{pmatrix} x_p \\ y_p \\ z_p \end{pmatrix} \tag{41}$$

First note that the units of (x_q, y_q) are independent of the units of (x_p, y_p, z_p) and are completely determined by the units of the focal length f. The latter could be expressed in milli-meter, pixels, or one can use normalized coordinates by setting $f = 1$. Secondly, the same projected point (x_q, y_q) is obtained by any point $(\alpha x_p, \alpha y_p, \alpha z_p)$ with $\alpha \in \mathbb{R}_0$.

With normal coordinates, the non-linear division by z_p prevents expressing the pinhole projection as a convenient matrix multiplication. We therefore go to augmented space and use homogeneous coordinates $(x_q, y_q, 1)$ and $(x_p, y_p, z_p, 1)$

$$\begin{pmatrix} x_q \\ y_q \\ z_q \\ 1 \end{pmatrix} \sim \mathbf{C} \begin{pmatrix} x_p \\ y_p \\ z_q \\ 1 \end{pmatrix} \tag{42}$$

where the equality $=$ has been replaced by the equivalence relation $\sim$. After any transformation a homogenisation of the coordinates is executed, which means dividing all coordinates with the fourth one. $\mathbf{C}$ is called the *camera matrix*, for which different versions circulate in the literature. For example:

$$\mathbf{C}_1 = \begin{pmatrix} f & 0 & 0 & 0 \\ 0 & f & 0 & 0 \\ 0 & 0 & f & 0 \\ 0 & 0 & -1 & 0 \end{pmatrix} \tag{43}$$

is the augmented equivalent of Eq. (41). The following version avoids mirroring the x- and y- axes of the projected reference frame:

$$\mathbf{C}_2 = \begin{pmatrix} f & 0 & 0 & 0 \\ 0 & f & 0 & 0 \\ 0 & 0 & f & 0 \\ 0 & 0 & 1 & 0 \end{pmatrix}. \tag{44}$$

Neither $\mathbf{C}_1$ and $\mathbf{C}_2$ are invertible, as the projection on a plane loses the original z_p coordinate. Alternatively, the following camera matrix retains this information, and is invertible:

$$\mathbf{C}_3 = \begin{pmatrix} f & 0 & 0 & 0 \\ 0 & f & 0 & 0 \\ 0 & 0 & f & 1 \\ 0 & 0 & 1 & 0 \end{pmatrix} \tag{45}$$

the resulting homogenised coordinate $z_q = 1 + 1/z_q$ coordinate is never used in practice, but simply guarantees invertibility of the matrix.

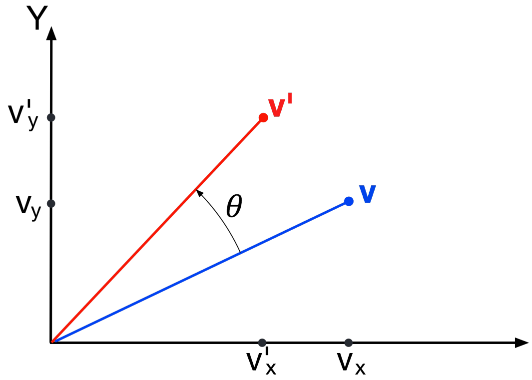

Figure 7: Rotation of a vector $\mathbf{v}$ around the z-axis $\mathbf{e}_z = \mathbf{u}$ by an angle θ to give the vector $\mathbf{v}'$. The z-axis of this right-handed reference frame is pointing towards the reader. The angle θ increases clockwise from vector $\mathbf{v}$ to vector $\mathbf{v}'$ when looking in the direction of the positive rotation axis.

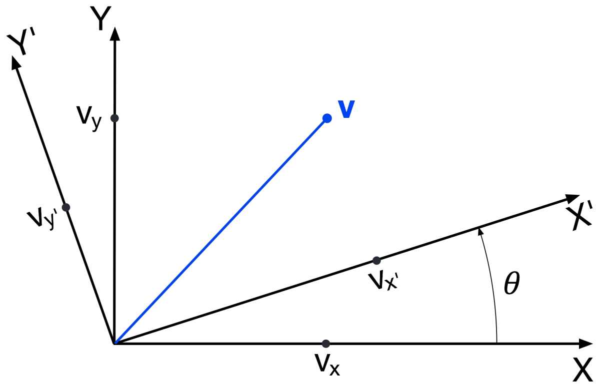

Figure 8: The components of the same vector $\mathbf{v}$ in the original reference frame XY and the rotated reference frame $X'Y'$. Looking in the direction of the positive $\mathbf{e}_{z'} = \mathbf{e}_z$ axis (which is pointing towards the reader), the rotation angle θ now increases clockwise from the X-axis to the X'-axis.

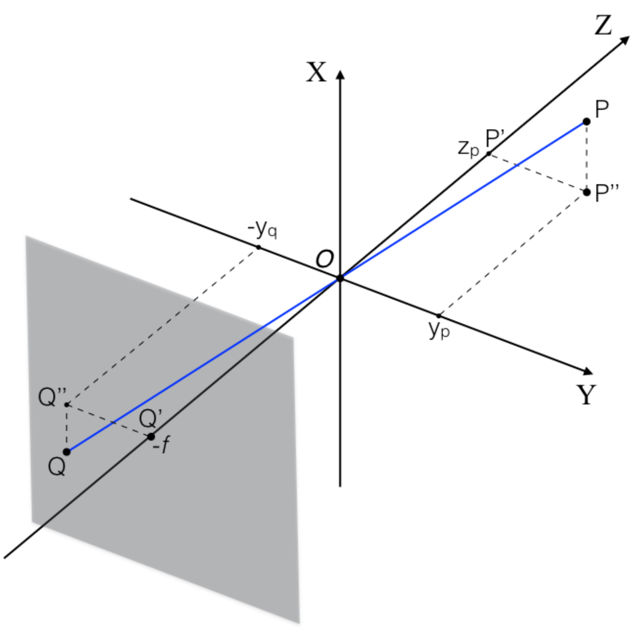

Figure 9: The standard pinhole camera model. The Z axis corresponds to the optical axis of the telescope. The origin of the reference frame is the pinhole O, the focal plane is located behind the pinhole at a distance f where f is the focal length of the camera. Light rays of a star at point P go through the pin hole O and are projected to a point Q on the focal plane.